AI in CAD: Learner's Guide

Strategic Application of Artificial Intelligence in CAD · SGSITS Mechanical, B.Tech 2026-27 · Supervisor Dr. B. R. Rawal · one overview page, then one page per presenter

The project in one line. You type "knuckle joint for 50 kN" and get back a joint that is sized from design-book formulas, checked against every failure mode, modelled in 3D, drawn on a first-angle A3 sheet and exported as STEP and DXF. AI only reads the sentence. All the engineering is plain, traceable code.

The pipeline

1. **Read.** A language model pulls out the joint type and the load, nothing else.
2. **Guard.** Any number the AI gives that isn't in the sentence is dropped and logged.
3. **Size.** The governing formula gives one size, rounded up to a standard size.
4. **Derive.** The other sizes follow from book proportions, each one tagged with its book and equation.
5. **Check.** Every failure mode (tension, shear, bending, crushing) at the real load. If one fails, the responsible size grows 1 mm and everything is re-checked.
6. **Model and draw.** CadQuery builds exact solids. FreeCAD, headless, draws the sheet.
7. **Cross-check.** gmsh plus CalculiX solve one part by FEA, as an independent check.

Numbers to quote

What	Value
Joints designed end to end	6
Failure checks, all satisfied	44; busiest per joint 71 to 79%
Knuckle at 50 kN: governing check	pin bending, 59.4 of 75 MPa (79%)
Knuckle sizes	rod D 35, pin d 45, eye d2 90, head d3 70, t 45, t1 30 mm
Bhandari Ex 4.5 (knuckle)	7 of 7 sizes reproduced exactly
Bhandari Ex 4.2 (cotter)	9 of 10 sizes (book uses FoS 4 for the cotter width)
Design factor	size at 1.25 times the load, check at the real load
Speed	about 0.03 ms per full knuckle design
Automated tests	301
Errors found in sources	3: NPTEL knuckle example fails its own eye-shear check by 23%; cotter spigot crushing check omitted; bolt preload printed 284d kN, should be 2840d N

Who presents what: Atishay = AI · Bhavya = AI in mechanical engineering · Ansh = machine design · Dharmendra = drawing and standards · Ankur = FEA and CAD geometry.

Atishay Jain

AI: language models, and why sizing is not ML

Core concepts

A **large language model (LLM)** is a neural network trained to predict the next **token**, which is a word piece (about 3-4 characters of English). It is very good at understanding messy language. It is not reliable at arithmetic, because it produces the most likely text and doesn't actually compute anything.

Hallucination is when the model states something fluent and wrong, e.g. a load nobody gave it. In engineering that is dangerous: a wrong number becomes a wrong part.

Rules vs LLM parsing. Rules (regex) are exact but brittle: "50 kilonewtons" or "5 tonne" can slip through. An LLM handles any phrasing but might invent things. We use both: regex first, then the LLM, and every LLM number is verified against the original text.

Why sizing is not ML. Machine learning learns patterns from data and gives no reason for its output. A joint's size must have a reason (formula, book, page) and must be exactly right at every load. Design formulas already exist, so learning them from data would be less accurate and harder to explain.

Text-to-CAD landscape. Commercial tools such as Zoo's Text-to-CAD and research models generate shapes straight from text. They produce geometry that looks right, but nothing proves it carries the load. Ours goes the other way: engineering first, then geometry.

How it shows up in our project

> *In `aicad/parse.py`, the function `_traceable(value, text)` checks that every number the AI returns appears in the user's sentence, allowing unit factors (x1000 for kN, g for tonne). If not, it is dropped with the note "AI gave ... which is not in the request; dropped". So the AI cannot inject a load. Sizes, checks and geometry never touch the AI.*

Viva questions

1. **Is this a trained model?**
   No. It is a rules engine from design books. AI only reads the request.
2. **What stops the AI hallucinating a number?**
   _traceable: any number not in the sentence is dropped and logged.
3. **Why not train a model to size joints?**
   Formulas already exist and are exact. ML would be approximate and unexplainable.
4. **What is a token?**
   A word piece the model reads and writes, roughly 3-4 characters.
5. **What if there's no API key or internet?**
   The regex parser still works for normal requests. The AI is a fallback for unusual phrasing.
6. **How is this different from text-to-CAD tools?**
   They generate shape. We generate a checked design, then the shape.

Bhavya Shrivastava

AI in mechanical engineering

Core concepts

Physics-informed neural networks (PINN). A normal network minimises the *data loss* (error against measurements). A PINN adds a *physics residual*, which is how far the output is from satisfying the governing equation (e.g. equilibrium). Loss = data loss + lambda x physics residual. Physics here is a **soft constraint**: penalised, but it can still be violated.

Surrogate models are fast approximations (neural nets, Gaussian processes) trained on expensive simulations, so a design can be evaluated in milliseconds instead of hours. They are used in optimisation loops.

Generative design and topology optimisation remove material where stress is low, which gives organic, light shapes (Fusion 360 generative design, SIMP method). The result must still be validated and made manufacturable.

Digital twins are a live virtual copy of a real machine, fed by sensors and used to predict wear and failure.

How it shows up in our project

> *aicad treats physics as a **hard constraint**: a design is only returned if every failure check passes. Nothing is penalised and traded off. That's the right choice for a few standard joints with known formulas. A PINN or a surrogate earns its place when there is no closed-form formula, e.g. a complex casting. So the honest pitch: we used AI where it is strong (language) and exact physics where a mistake breaks a part.*

Viva questions

1. **What is a PINN?**
   A network trained on data loss plus a physics-equation residual, so physics is a soft constraint.
2. **Soft vs hard constraint?**
   Soft = penalised and can be violated. Hard = must hold. aicad uses hard checks.
3. **Why didn't you use a PINN?**
   The joint formulas are closed-form and exact. A PINN would only approximate them.
4. **What is a surrogate model?**
   A fast learned stand-in for an expensive simulation, used inside optimisation.
5. **Topology optimisation in one line?**
   Remove material where stress is low until the stiffness target is met.
6. **Where would AI help next?**
   Surrogates for FEA on unlisted shapes, and generative design for custom brackets.

Ansh Yadav

Machine design of joints

Core concepts

Failure modes. Each part can fail by tension (pulled apart), shear (sliced across), bending (pin bowed between fork plates) or crushing (bearing pressure on contact faces). Design means sizing so that each stress stays under its allowable value.

Allowable stress = yield strength / factor of safety: sigma_t = Syt / FoS. For shear, tau = 0.5 Syt / FoS (maximum shear stress theory).

Knuckle, Bhandari eq 4.26. Rod D from tension. Pin d from shear *and* bending, whichever is larger. Then: eye outer d0 = 2d, fork thickness a = 0.75D, eye thickness b = 1.25D, pin head 1.5d, enlarged rod D1 = 1.1D, clearance x = 10 mm.

Socket and spigot cotter. Rod d, cotter thickness t = 0.31d, collar d3 = 1.5d, socket collar d4 = 2.4d, a = c = 0.75d, cotter width b = 1.6d, collar thickness t1 = 0.45d, cotter taper 1 in 32 (self-locking). Spigot and socket diameters are solved from their own tension equations.

How it shows up in our project

> *At 50 kN the rod needs 32.57 mm, so D = 35. The pin needs 25.75 mm for shear but 44.85 mm for bending, so bending governs and d = 45. Then d2 = 90, d3 = 70, t = 45, t1 = 30. The busiest check is pin bending at 79%. Bhandari Ex 4.5 is reproduced 7/7 sizes.*

Viva questions

1. **Why size the pin for bending too?**
   Shear alone under-sizes it. Here bending needs 44.85 mm vs 25.75 for shear.
2. **Why is the cotter tapered 1 in 32?**
   A small taper is self-locking under friction but still drives out with a hammer.
3. **What is allowable shear stress?**
   0.5 Syt / FoS, from the maximum shear stress theory.
4. **Why do checks sit at 71-79%, not 100%?**
   Sized at 1.25 x load, checked at the real load, plus rounding up.
5. **Knuckle vs cotter joint?**
   Knuckle allows angular movement (pin). Cotter is rigid, for axial load only.
6. **What happens if a check fails?**
   The responsible size grows 1 mm and every check re-runs until all pass.

Dharmendra Gupta

Engineering drawing and standards

Core concepts

First-angle projection (Indian standard, SP 46 / IS 10711): the object sits between observer and plane, so the top view goes *below* the front view and the left view goes on the *right*. The symbol is a truncated cone shown with its circle views.

Sectioning. A cutting plane (marked A-A, a chain line thick at the ends, arrows show viewing direction) reveals the inside. Cut faces are hatched with thin lines at 45 degrees, and adjacent parts get opposite or different-spaced hatching.

Not hatched lengthwise: solid pins, bolts, nuts, keys, shafts, cotters, rivets and ribs. Cutting them along their length gives no information, so they're drawn whole.

Centre lines (thin chain) on every axis and hole. A **parts list** gives each part's number, name, material and quantity, linked by balloons.

Tolerances, fits and GD&T. Fits (e.g. H7/g6 for a pin in a hole) set the clearance. GD&T controls form and position (flatness, perpendicularity, concentricity) against datums.

How it shows up in our project

> *FreeCAD runs headless and produces an A3 first-angle sheet: the front view as sectioned A-A with hatching, the pin and cotter left unhatched, centre lines, dimensions, a title block and a parts list, plus a DXF.* ***Not implemented yet:*** *tolerances, fits and GD&T. Faculty raised this on 16 October, so say it before they do.*

Viva questions

1. **Why first-angle?**
   It is the Indian standard SP 46 that we're taught. Top view below, left view on the right.
2. **Why is the pin not hatched?**
   Solid pins, bolts, keys, shafts and cotters aren't sectioned lengthwise.
3. **How do you show the cutting plane?**
   A chain line, thick at the ends, labelled A-A, with arrows for the viewing direction.
4. **What fit would the knuckle pin use?**
   A clearance fit like H9/d9 so it can rotate. Not yet on our drawing.
5. **What is GD&T?**
   Symbols controlling form and position, e.g. perpendicularity to a datum.
6. **What does the parts list carry?**
   Part number, name, material and quantity, linked to balloons.

Ankur Dodke

FEA and CAD geometry

Core concepts

FEM basics. Split the part into small elements (a mesh), assume a simple displacement field in each, and assemble a stiffness equation K u = F. Solve for displacement u, then compute strain and stress.

Meshing (gmsh). Tetrahedral elements, finer where stress changes fast (holes, fillets). Second-order (10-node) tets are much more accurate than linear ones.

Convergence and singularities. Refine the mesh until the result stops changing: that's convergence. At a sharp re-entrant corner or a point load the stress keeps rising with every refinement. That is a *singularity*, a modelling artefact, not real stress, so don't report it.

CalculiX is an open-source solver that reads Abaqus-style input. **B-rep** (boundary representation) stores a solid as exact faces, edges and vertices. **OpenCASCADE** is the kernel and **CadQuery** is the Python API on top of it. **STEP AP214** is the neutral format that carries the exact geometry into NX or SolidWorks.

How it shows up in our project

> *Validation on a plain bar: hand formula 31.83 MPa vs FE 32.22 MPa, so +1.2%, and the solver is trusted. Knuckle eye rod: hand formula 52.0 vs FE 57.9 MPa, +11.3%. The gap is stress concentration that the formula ignores. Both were measured on the September geometry, and CalculiX runs only on the server. If the mesh won't settle, the software says NOT CONVERGED rather than giving a number.*

Viva questions

1. **Why validate on a plain bar first?**
   Its exact answer is known, so +1.2% proves the solver setup.
2. **Why is FE 11% higher on the eye?**
   Stress concentration at the hole, which the textbook formula doesn't model.
3. **What is a singularity?**
   A sharp corner or point load where stress rises forever with refinement. It's an artefact.
4. **How do you know it converged?**
   The result changes negligibly between mesh refinements. Otherwise it's flagged NOT CONVERGED.
5. **B-rep vs mesh?**
   B-rep is exact surfaces for CAD and STEP. A mesh is an approximation for FEA.
6. **Why STEP AP214?**
   A neutral exact-geometry format that NX, SolidWorks and Fusion all open as an assembly.